

CBSE Class 12 Chemistry — Half-Yearly Predicted Paper (Set 10 of 10)

Section A: 1-Mark Questions (MCQs / Assertion-Reason)

Q1. Which of the following concentration terms is independent of temperature? (a) Molarity (b) Molality (c) Normality (d) Formality [Solutions] [PYQ 2017] [Confidence: Very High]

Q2. The standard reduction potentials at **298 K** for the following half cells are given: $Zn^{2+}/Zn = -0.76 V$, $Cr^{3+}/Cr = -0.74 V$, $H^+/H_2 = 0.00 V$, $Fe^{3+}/Fe^{2+} = +0.77 V$. Which of the following is the strongest reducing agent? (a) $Zn(s)$ (b) $Cr(s)$ (c) $H_2(g)$ (d) $Fe^{2+}(aq)$ [Electrochemistry] [Practice] [Confidence: High]

Q3. A first-order reaction has a specific rate constant $k = 1.0 \times 10^{-3} s^{-1}$. What is its half-life period? (a) **69.3 s** (b) **693 s** (c) **0.693 s** (d) **6.93 s** [Chemical Kinetics] [PYQ 2018] [Confidence: Very High]

Q4. The actinoid contraction is slightly greater from element to element than the lanthanoid contraction. This is primarily due to: (a) Poor shielding of the 5f electrons compared to the 4f electrons. (b) Poor shielding of the 4f electrons compared to the 5f electrons. (c) Greater effective nuclear charge in lanthanoids. (d) The radioactive nature of actinoids. [The d- and f-Block Elements] [PYQ 2019] [Confidence: Very High]

Q5. What is the oxidation state of iron (Fe) in the brown ring complex $[Fe(H_2O)_5(NO)]SO_4$? (a) +1 (b) +2 (c) +3 (d) 0 [Coordination Compounds] [Practice] [Confidence: High]

Q6. Which of the following is an ambident nucleophile? (a) OH^- (b) CN^- (c) NH_3 (d) H_2O [Haloalkanes and Haloarenes] [PYQ 2015] [Confidence: Very High]

Q7. Which of the following compounds will NOT form a yellow precipitate on heating with an alkaline solution of iodine? (a) $CH_3CH(OH)CH_3$ (b) CH_3CH_2OH (c) CH_3OH (d) $CH_3CH_2CH(OH)CH_3$ [Alcohols, Phenols and Ethers] [PYQ SQP 2021] [Confidence: Very High]

Q8. Which of the following compounds does NOT give a positive silver mirror test with Tollen's reagent? (a) Formaldehyde ($HCHO$) (b) Acetaldehyde (CH_3CHO) (c) Acetone (CH_3COCH_3) (d) Benzaldehyde (C_6H_5CHO) [Aldehydes, Ketones and Carboxylic Acids] [Practice] [Confidence: Very High]

Q9. If the molal depression constant (K_f) for water is $1.86 K kg mol^{-1}$, what is the estimated freezing point of a **0.1 m** aqueous solution of $NaCl$ (assuming complete dissociation)? (a) $-0.186^\circ C$ (b) $-0.372^\circ C$ (c) $+0.372^\circ C$ (d) $-0.558^\circ C$ [Solutions] [Practice] [Confidence: High]

Q10. The rusting of iron is fundamentally an electrochemical process which is heavily catalyzed by the presence of: (a) Zn (b) OH^- (c) O_2 (d) H^+ [Electrochemistry] [PYQ 2016] [Confidence: Very High]

Q11. For a chemical reaction $A \rightarrow B$, the rate of reaction quadruples when the concentration of reactant A is doubled. The overall order of this reaction is: (a) 0 (b) 1 (c) 2 (d) 3 [Chemical Kinetics] [PYQ 2015] [Confidence: Very High]

Q12. The specific reagents heavily utilized in the Sandmeyer reaction to convert a diazonium salt to an aryl halide are: (a) $CuCl/HCl$ or $CuBr/HBr$ (b) Cu powder/ HCl (c) Zn/HCl (d) Fe/HCl [Haloalkanes and Haloarenes] [PYQ 2014] [Confidence: Very High]

Directions for Q13 to Q16: A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): The coordination complex $[Fe(CN)_6]^{3-}$ is weakly paramagnetic. Reason (R): Cyanide (CN^-) acts as a weak field ligand, leaving 5 unpaired electrons in the 3d orbitals of iron. [Coordination Compounds] [Practice] [Confidence: High]

Q14. Assertion (A): The $C - O - C$ bond angle in ethers is slightly greater than the normal tetrahedral angle ($109^\circ 28'$). Reason (R): There is a strong repulsive interaction between the two bulky alkyl groups attached to the oxygen atom. [Alcohols, Phenols and Ethers] [PYQ 2019] [Confidence: Very High]

Q15. Assertion (A): Transition metals possess remarkably high enthalpies of atomisation. Reason (R): They exhibit strong metallic bonding due to the active participation of a large number of unpaired d-electrons. [The d- and f-Block Elements] [PYQ 2018] [Confidence: Very High]

Q16. Assertion (A): Carboxylic acids do not give the characteristic nucleophilic addition reactions of the carbonyl group (like with 2,4-DNP or HCN). Reason (R): The carbonyl carbon in carboxylic acids is significantly less electrophilic because it

is involved in resonance with the lone pair of electrons on the adjacent hydroxyl oxygen. [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2017] [Confidence: Very High]

Section B: 2-Mark Questions (Short Answer)

Q17. A mixture of ethanol and acetone shows a positive deviation from Raoult's law. Explain the molecular-level reason behind this observation and state the sign of ΔH_{mix} for this solution. [Solutions] [PYQ 2019] [Confidence: Very High]

Q18. The limiting molar conductivity (Λ_m°) of water cannot be determined experimentally by simply extrapolating a plot of Λ_m vs $\sqrt{c}$. Suggest a reliable method using Kohlrausch's law to determine the Λ_m° of water, providing the appropriate mathematical expression. [Electrochemistry] [Practice / PYQ Theme] [Confidence: High]

Q19. Show mathematically that for a zero-order reaction, the half-life period ($t_{1/2}$) is directly proportional to the initial concentration of the reactant ($[R]_0$). [Chemical Kinetics] [PYQ 2016] [Confidence: Very High]

Q20. Using Valence Bond Theory (VBT), predict the state of hybridization and the magnetic behavior (paramagnetic or diamagnetic) of the complex ion $[FeF_6]^{3-}$. (Given: Atomic number of Fe = 26). [Coordination Compounds] [CBSE Term-2 2022] [Confidence: Very High]

Q21. Arrange the following alkyl halides in the strictly increasing order of their reactivity towards an S_N2 substitution mechanism and briefly justify your answer based on steric hindrance: CH_3Cl , CH_3Br , CH_3CH_2Cl , $(CH_3)_2CHCl$. [Haloalkanes and Haloarenes] [Practice] [Confidence: High]

Section C: 3-Mark Questions (Short Answer)

Q22. Calculate the exact mass of an unknown non-volatile solute (molar mass = 256 g mol^{-1}) that must be dissolved in 75 g of benzene to lower its freezing point by precisely 0.48 K . (Given: The molal depression constant K_f for benzene = $5.12 \text{ K kg mol}^{-1}$). [Solutions] [PYQ 2015] [Confidence: Very High]

Q23. An electrochemical cell in which the following redox reaction occurs: $2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(s)$ has a standard cell potential (E_{cell}°) of 0.236 V at 298 K . Calculate the standard Gibbs free energy change (ΔG°) and the equilibrium constant (K_c) for the cell reaction. (Given: $1F = 96500 \text{ C mol}^{-1}$, $\frac{2.303RT}{F} = 0.0591 \text{ V}$). [Electrochemistry] [PYQ 2019] [Confidence: Very High]

Q24. The specific rate constant (k) for a first-order chemical reaction is precisely 60 s^{-1} . How much time will it take to reduce the initial concentration of the reactant to exactly $1/16^{th}$ of its original value? [Chemical Kinetics] [PYQ 2017 / Practice] [Confidence: High]

Q25. Provide distinct chemical reasons to account for the following observations: (i) The Cu^+ ion is diamagnetic and colourless, whereas the Cu^{2+} ion is paramagnetic and coloured in an aqueous solution. (ii) Zinc, Cadmium, and Mercury are strictly not considered transition elements. (iii) The Actinoid contraction is greater from element to element than the Lanthanoid contraction. [The d- and f-Block Elements] [PYQ 2023 / 2018] [Confidence: Very High]

Q26. Write the precise chemical equations and necessary reagents to carry out the following organic conversions: (i) Phenol to 2,4,6-Trinitrophenol (Picric acid) (ii) Propan-1-ol to 2-Bromopropane (iii) Anisole to p-Bromoanisole [Alcohols, Phenols and Ethers] [PYQ 2016 / 2020] [Confidence: Very High]

Q27. Write the balanced chemical equations to clearly illustrate the following named organic reactions: (i) Etard reaction (ii) Clemmensen reduction (iii) Hell-Volhard-Zelinsky (HVZ) reaction [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2015 / 2018] [Confidence: Very High]

Q28. (i) Explain the stereochemical outcome of an S_N1 reaction with a suitable optically active alkyl halide example. (1.5 Marks) (ii) Explain the stereochemical outcome of an S_N2 reaction, defining the term "Walden inversion". (1.5 Marks) [Haloalkanes and Haloarenes] [PYQ 2019] [Confidence: Very High]

Section D: 4-Mark Questions (Case-Study)

Q29. Case Study: Osmosis and Reverse Osmosis Osmosis is the spontaneous flow of solvent molecules from a region of lower solute concentration to a region of higher solute concentration through a semi-permeable membrane. The pressure that must be applied to the solution to just stop this flow is called the osmotic pressure (π). However, if a pressure strictly larger than the osmotic pressure is applied to the solution side, the direction of osmosis is reversed. This phenomenon,

called reverse osmosis (RO), is heavily utilized in countries facing water scarcity for the desalination of seawater to obtain fresh, potable drinking water. (a) Define the term 'reverse osmosis'. (1 Mark) (b) Name the chemical nature of the semi-permeable membrane widely used in commercial reverse osmosis plants for desalination. (1 Mark) (c) Calculate the osmotic pressure (π) of a **0.01 M** aqueous solution of Barium Chloride (BaCl_2) at **300 K**, assuming it undergoes complete dissociation in water. (Given: $R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$). (2 Marks) [Solutions] [CBSE SQP Theme / Practice] [Confidence: High]

Q30. Case Study: Nucleophilic Addition in Carbonyl Compounds Aldehydes and ketones are highly susceptible to nucleophilic addition reactions due to the polar nature of the carbon-oxygen double bond. The carbonyl carbon is electrophilic (electron-deficient) and is readily attacked by a nucleophile. However, aldehydes are generally much more reactive than ketones towards these reactions. This difference in reactivity is primarily attributed to two factors: steric hindrance (the presence of two bulky alkyl groups in ketones physically blocks the nucleophile) and electronic effects (the electron-donating +I effect of two alkyl groups in ketones reduces the electrophilicity of the carbonyl carbon). (a) Write the exact chemical structure of the product formed when acetaldehyde (CH_3CHO) undergoes nucleophilic addition with Hydrogen Cyanide (HCN). (1 Mark) (b) Chemically, why is benzaldehyde less reactive than propanal towards nucleophilic addition reactions? (1 Mark) (c) Arrange the following compounds in the strictly increasing order of their reactivity towards addition of HCN : Benzaldehyde, p-Tolualdehyde ($p\text{-CH}_3\text{-C}_6\text{H}_4\text{-CHO}$), p-Nitrobenzaldehyde ($p\text{-NO}_2\text{-C}_6\text{H}_4\text{-CHO}$), Acetophenone. (2 Marks) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2020 / Practice] [Confidence: Very High]

Section E: 5-Mark Questions (Long Answer)

Q31. (a) Define the term *Molar Conductivity* (Λ_m). Explain in detail how the molar conductivity of a weak electrolyte (like acetic acid) varies with a decrease in concentration (dilution). (2 Marks) (b) The specific rate constant (k) for a first-order chemical reaction is found to be such that the reaction is exactly **20%** complete in 10 minutes. Calculate the exact time taken for this reaction to reach **75%** completion. (Given: $\log(10) = 1.000$, $\log(8) = 0.903$, $\log(4) = 0.602$). (3 Marks) [Electrochemistry / Chemical Kinetics] [PYQ 2018 / 2016] [Confidence: Very High]

Q32. (a) Using Valence Bond Theory (VBT), deduce the state of hybridization, the resulting geometry, and the magnetic behavior of the complex ion $[\text{Co}(\text{NH}_3)_6]^{3+}$. (Given: Atomic number of Co = 27). (3 Marks) (b) Transition metals possess an extraordinary ability to form a vast array of complex coordination compounds. Provide two distinct fundamental chemical reasons to strongly justify this unique capability. (2 Marks) [Coordination Compounds / The d- and f-Block Elements] [PYQ 2015 / 2019] [Confidence: Very High]

Q33. An unknown organic compound **A** with the molecular formula $\text{C}_8\text{H}_8\text{O}$ forms an orange-red precipitate with 2,4-DNP reagent and also gives a positive yellow precipitate with the iodoform test. However, it completely fails to reduce Tollens' reagent. When compound **A** is subjected to vigorous oxidation with strongly acidic KMnO_4 , it yields a stable carboxylic acid **B** with the molecular formula $\text{C}_7\text{H}_6\text{O}_2$. Compound **B** on treatment with Thionyl chloride (SOCl_2) rapidly yields compound **C**. (a) Deduce the exact structures and standard IUPAC names of compounds **A**, **B**, and **C**. (3 Marks) (b) Write the fully balanced chemical equation for the iodoform reaction of compound **A**. (1 Mark) (c) What happens when compound **B** is treated with a mixture of concentrated HNO_3 and concentrated H_2SO_4 ? Write the structure of the product formed. (1 Mark) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2016 / 2023] [Confidence: Very High]

Answer Key

Q1. (b) Molality. (Volume changes with temperature, but mass does not. Since molality is based on the mass of the solvent, it remains strictly independent of temperature). **Q2.** (a) $\text{Zn}(s)$. (The lower the standard reduction potential, the stronger the tendency to get oxidized and thus act as a reducing agent. Zn has the lowest potential at -0.76 V). **Q3.** (b) **693 s.** ($t_{1/2} = 0.693/k = 0.693/1.0 \times 10^{-3} = 693 \text{ seconds}$). **Q4.** (a) Poor shielding of the 5f electrons compared to the 4f electrons. (This causes a stronger inward pull by the nucleus on the outermost electrons). **Q5.** (a) +1. (In the brown ring complex, NO acts as NO^+ . Water is neutral. $x + 0 + 1 = 2$ (charge on complex sphere matching sulfate). So $x = +1$). **Q6.** (b) CN^- . (It possesses two nucleophilic centers: Carbon and Nitrogen, capable of attacking through either). **Q7.** (c) CH_3OH . (Methanol lacks the $\text{CH}_3\text{-CH(OH)-}$ group required to give a positive iodoform test). **Q8.** (c) Acetone (CH_3COCH_3). (Tollen's reagent is a mild oxidizing agent that can oxidize aldehydes but not ketones). **Q9.** (b) -0.372°C . ($\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$, so $i = 2$. $\Delta T_f = i \cdot K_f \cdot m = 2 \times 1.86 \times 0.1 = 0.372^\circ\text{C}$. Freezing point = $0 - 0.372 = -0.372^\circ\text{C}$). **Q10.** (d) H^+ . (The presence of H^+ ions from carbonic acid or atmospheric moisture accelerates

the reduction of oxygen at the cathodic sites during rusting). **Q11.** (c) 2. (Rate $\propto [A]^n$. $4 = 2^n \implies n = 2$). **Q12.** (a) ***CuCl/HCl*** or ***CuBr/HBr***. (Using cuprous halides. Cu powder is used in the Gattermann reaction). **Q13.** (c) A is true but R is false. (Cyanide (CN^-) is a *strong* field ligand, which forces electron pairing, leaving only 1 unpaired electron in Fe^{3+} , making it *weakly* paramagnetic). **Q14.** (a) Both A and R are true and R is the correct explanation of A. **Q15.** (a) Both A and R are true and R is the correct explanation of A. **Q16.** (a) Both A and R are true and R is the correct explanation of A. (Resonance stabilization strictly decreases the electrophilicity of the carbonyl carbon).

Q17. Ethanol molecules are strongly hydrogen-bonded. When acetone is added, its molecules insert between ethanol molecules, breaking some of the strong H-bonds. The new interactions (ethanol-acetone) are weaker than the pure interactions, making it easier for molecules to escape into the vapor phase (increasing vapor pressure). For positive deviations, $\Delta H_{mix} > 0$ (endothermic). **Q18.** According to Kohlrausch's law, $\Lambda_m^\circ(H_2O) = \lambda_m^\circ(H^+) + \lambda_m^\circ(OH^-)$. We can determine this using strong electrolytes: $\Lambda_m^\circ(H_2O) = \Lambda_m^\circ(HCl) + \Lambda_m^\circ(NaOH) - \Lambda_m^\circ(NaCl)$. **Q19.** For a zero-order reaction, the integrated rate equation is $[R] = -kt + [R]_0$. At half-life ($t = t_{1/2}$), the concentration $[R] = \frac{[R]_0}{2}$. Substituting this: $\frac{[R]_0}{2} = -k(t_{1/2}) + [R]_0 \implies k(t_{1/2}) = [R]_0 - \frac{[R]_0}{2} = \frac{[R]_0}{2}$. Thus, $t_{1/2} = \frac{[R]_0}{2k}$. Since $2k$ is constant, $t_{1/2} \propto [R]_0$. **Q20.** Fe ($Z=26$) is $3d^6 4s^2$. In $[FeF_6]^{3-}$, Fe is in the +3 oxidation state ($3d^5$). Fluoride (F^-) is a weak field ligand, so pairing of electrons does not occur against Hund's rule. The complex utilizes outer 4d orbitals. Hybridization: $sp^3 d^2$ (Outer orbital complex). Magnetic Behavior: Paramagnetic (due to the presence of 5 unpaired electrons). **Q21.** $(CH_3)_2CHCl < CH_3CH_2Cl < CH_3Cl < CH_3Br$. Reason: S_N2 reactivity strictly decreases with increasing steric hindrance. Isopropyl chloride is 2° , ethyl is 1° , and methyl is unhindered. Bromide is a better leaving group than chloride, making CH_3Br the most reactive.

Q22. $\Delta T_f = K_f \times m = K_f \times \frac{W_B \times 1000}{M_B \times W_A}$. $0.48 = 5.12 \times \frac{W_B \times 1000}{256 \times 75}$. $0.48 = 5.12 \times \frac{W_B \times 1000}{19200} \implies W_B = \frac{0.48 \times 19200}{5.12 \times 1000} = \frac{9216}{5120} = 1.8$ g. **Q23.** $n = 2$ (since 2 electrons are transferred: $2Fe^{3+} \rightarrow 2Fe^{2+}$). $\Delta G^\circ = -nFE_{cell}^\circ = -2 \times 96500 \text{ C mol}^{-1} \times 0.236 \text{ V} = -45,548 \text{ J mol}^{-1}$ or $-45.548 \text{ kJ mol}^{-1}$. Using $E_{cell}^\circ = \frac{0.0591}{n} \log K_c \implies 0.236 = \frac{0.0591}{2} \log K_c \implies \log K_c = \frac{0.472}{0.0591} \approx 7.986$. $K_c = \text{Antilog}(7.986) = 9.68 \times 10^7$. **Q24.** For a 1st order reaction, $t = \frac{2.303}{k} \log \left(\frac{[R]_0}{[R]} \right)$. $[R] = \frac{[R]_0}{16}$. $t = \frac{2.303}{60} \log(16) = \frac{2.303}{60} \log(2^4) = \frac{2.303 \times 4 \times 0.301}{60} = \frac{2.772}{60} = 0.0462 \text{ s}$. (Alternatively, $1/16$ means 4 half-lives. $t_{1/2} = 0.693/60 = 0.01155 \text{ s}$. Total time = $4 \times 0.01155 = 0.0462 \text{ s}$). **Q25.** (i) Cu^+ has a fully filled $3d^{10}$ configuration with no unpaired electrons (diamagnetic/colourless). Cu^{2+} has a $3d^9$ configuration with 1 unpaired electron, allowing d-d transitions (paramagnetic/coloured). (ii) Transition elements are defined as having an incompletely filled d-subshell. Zn, Cd, and Hg have completely filled d-orbitals (d^{10}) in their ground state and common oxidation states. (iii) The 5f electrons in actinoids possess poorer shielding effect than the 4f electrons in lanthanoids, leading to a much stronger inward pull by the effective nuclear charge. **Q26.** (i) Phenol

$+3HNO_3$ (conc.) $\xrightarrow{\text{Conc. } H_2SO_4}$ 2,4,6-Trinitrophenol (Picric acid) + $3H_2O$. (ii) $CH_3CH_2CH_2OH \xrightarrow{PBr_3} CH_3CH_2CH_2Br \xrightarrow{\text{Alc. KOH}, \Delta} CH_3CH=CH_2 \xrightarrow{HBr} CH_3CH(Br)CH_3$. (iii) Anisole ($C_6H_5OCH_3$) + $Br_2 \xrightarrow{\text{Ethanoic acid}}$ p-Bromoanisole (Major) + o-Bromoanisole (Minor). **Q27.** (i) Etard: Toluene ($C_6H_5CH_3$) + $CrO_2Cl_2 \xrightarrow{CS_2} \text{Complex} \xrightarrow{H_3O^+}$ Benzaldehyde (C_6H_5CHO). (ii) Clemmensen: Ketone/Aldehyde ($R_2C=O$) $\xrightarrow{Zn(Hg), \text{conc. } HCl}$ Alkane (R_2CH_2) + H_2O . (iii) HVZ: Carboxylic acid with α -H (e.g., CH_3COOH) + $Cl_2 \xrightarrow{\text{Red P}}$ $Cl-CH_2-COOH + HCl$. **Q28.** (i) In an S_N1 mechanism, the reaction proceeds via a planar carbocation intermediate. The nucleophile can attack this planar face from either the front or the back with equal probability. Consequently, an optically active substrate yields an equimolar mixture of (+) and (-) enantiomers, leading to strict **racemization**. (ii) In an S_N2 mechanism, the nucleophile attacks the carbon atom from the side strictly opposite to the leaving group. This forces the remaining bonds to flip inside out, resulting in the complete **inversion of configuration**, known as Walden inversion.

Q29. (a) The process where solvent flows from a solution (higher concentration) to the pure solvent (lower concentration) through a semi-permeable membrane when a pressure strictly larger than the osmotic pressure is applied on the solution side. (b) Cellulose acetate. (c) $BaCl_2 \rightarrow Ba^{2+} + 2Cl^-$. Van't Hoff factor $i = 3$. $\pi = i \times C \times R \times T = 3 \times 0.01 \text{ mol L}^{-1} \times 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1} \times 300 \text{ K}$. $\pi = 0.03 \times 24.63 = 0.7389 \text{ atm}$.

Q30. (a) Acetaldehyde cyanohydrin: $CH_3-CH(OH)-CN$. (b) In benzaldehyde, the +R (resonance) effect of the benzene ring delocalizes electron density onto the carbonyl carbon, heavily decreasing its electrophilicity compared to the aliphatic propanal. (c) Acetophenone < p-Tolualdehyde < Benzaldehyde < p-Nitrobenzaldehyde. (Sterically, ketones are least reactive. Among aldehydes, EDG (methyl) decreases reactivity, while EWG (nitro) increases reactivity by intensifying the positive charge on the carbonyl carbon).

Q31. (a) Molar conductivity (Λ_m) is the conducting power of all the ions produced by dissolving one mole of an electrolyte in a solution. For a weak electrolyte, Λ_m increases steeply on dilution, especially at very low concentrations. This is because dilution significantly increases its degree of dissociation (α), thereby drastically increasing the total number of mobile ions in the solution. (b) 20% complete means 80% is left. $[R] = 0.8[R]_0$.

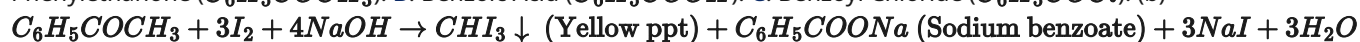
$$k = \frac{2.303}{10} \log \left(\frac{100}{80} \right) = 0.2303 \times (\log 10 - \log 8) = 0.2303 \times (1.000 - 0.903) = 0.2303 \times 0.097 = 0.0223 \text{ min}^{-1}.$$

$$\text{Time for 75\% completion (25\% left): } t = \frac{2.303}{0.0223} \log \left(\frac{100}{25} \right) = \frac{2.303}{0.0223} \log(4) = 103.27 \times 0.602 = 62.17 \text{ minutes. Q32. (a)}$$

Co is $3d^7 4s^2$. Co^{3+} is $3d^6$. Ammonia (NH_3) acts as a strong field ligand for Co^{3+} , forcing the complete pairing of the 3d electrons, which leaves two 3d orbitals completely empty. Hybridization: d^2sp^3 (Inner orbital complex). Geometry:

Octahedral. Magnetic behavior: Diamagnetic (0 unpaired electrons). (b) 1. Transition metal ions possess a high nuclear charge and remarkably small ionic sizes, drawing ligands closely. 2. They have completely vacant d-orbitals of appropriate energy perfectly suited to accept lone pairs of electrons donated by the ligands. **Q33. (a) A:** Acetophenone / 1-

Phenylethanone ($C_6H_5COCH_3$). **B:** Benzoic Acid (C_6H_5COOH). **C:** Benzoyl Chloride (C_6H_5COCl). (b)



. (c) The carboxyl group ($-COOH$) is strongly electron-withdrawing and meta-directing. Treatment of Benzoic acid with conc. HNO_3 and conc. H_2SO_4 (Nitration) yields **m-Nitrobenzoic acid**.

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